

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Units 1-6 & 9 (Measurement, Kinematics, Dynamics-I & II, Pressure & Deformation, Work & Energy [till 6.4], Nature of Science)

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions $\times$ 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions $\times$ 5 marks each. Answer *either* the main question *or* the OR alternative.
5. Use $g = 10 \text{ m/s}^2$ in numerical problems unless stated otherwise. Scientific calculator allowed.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	The prefix nano represents a multiplier of:	10^{-6}	10^{-9}	10^{-12}	10^9	(A)(B)(C)(D)
2	A screw gauge has a pitch of 1 mm and 100 divisions on its circular scale. Its least count is:	0.1 mm	0.01 mm	0.001 mm	1 mm	(A)(B)(C)(D)
3	The area under a velocity-time graph represents the:	Acceleration	Distance covered	Speed	Force applied	(A)(B)(C)(D)
4	A car starts from rest and reaches 30 m/s in 6 s. Its acceleration is:	5 m/s^2	0.2 m/s^2	180 m/s^2	36 m/s^2	(A)(B)(C)(D)
5	The SI unit of impulse is:	N·m	N·s	kg/s	J	(A)(B)(C)(D)
6	A body of mass 2 kg moves at 5 m/s. Its momentum is:	$10 \text{ kg}\cdot\text{m/s}$	$2.5 \text{ kg}\cdot\text{m/s}$	$25 \text{ kg}\cdot\text{m/s}$	$7 \text{ kg}\cdot\text{m/s}$	(A)(B)(C)(D)
7	The centripetal force on a body moving in a circle is always directed:	Along the tangent	Away from the centre	Towards the centre	Opposite to the motion	(A)(B)(C)(D)
8	For the same pair of surfaces, which type of friction is generally the smallest?	Static friction	Sliding friction	Rolling friction	Limiting friction	(A)(B)(C)(D)
9	A force of 200 N acts on an area of 0.5 m^2 . The pressure produced is:	100 Pa	400 Pa	40 Pa	0.0025 Pa	(A)(B)(C)(D)
10	Work done by a force is zero when the angle between the force and the displacement is:	0°	45°	90°	180°	(A)(B)(C)(D)
11	A 2 kg body moves at 3 m/s. Its kinetic energy is:	6 J	9 J	18 J	3 J	(A)(B)(C)(D)
12	A concise statement, often expressed mathematically, that summarizes a relationship always observed in nature is called a:	Hypothesis	Theory	Law	Experiment	(A)(B)(C)(D)

SECTION B — Short Answer Questions ($11 \times 3 = 33$ Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Define pitch and least count of a screw gauge. A screw gauge has a pitch of 1 mm and 50 divisions on its circular scale. Calculate its least count.	3	OR	Define significant figures . State how many significant figures are in (a) 0.0067 m and (b) 30500 J, and round 4.5678 to three significant figures.	3
(ii)	Write 0.000045 m and 6,400,000 m in scientific notation . State one reason why scientific notation is useful in physics.	3	OR	Write the SI prefix and symbol for each multiplier: 10^6 , 10^{-3} , 10^{-9} . Then convert 5 km into metres.	3
(iii)	A cyclist travels 300 m due east in 20 s, then 100 m due west in 10 s. Calculate (a) the total distance, (b) the net displacement, and (c) the average speed for the whole trip.	3	OR	Define average speed and instantaneous speed . A train covers 180 km in 2 hours. Find its average speed in m/s.	3

(iv)	State Newton's second law of motion . A force of 24 N acting on a body produces an acceleration of 3 m/s ² . Calculate the mass of the body.	3	OR	Define inertia . Using the idea of inertia, explain why passengers lurch forward when a moving bus brakes suddenly.	3
(v)	State the law of conservation of momentum . A 60 kg skater, initially at rest, throws a 2 kg ball horizontally at 9 m/s. Find the recoil speed of the skater.	3	OR	A 0.15 kg cricket ball moving at 20 m/s is struck and returns at 20 m/s in the opposite direction. Calculate the change in momentum and the impulse applied to the ball.	3
(vi)	Define torque (moment of force) and write its formula. A force of 50 N is applied at the end of a wrench 0.4 m long, perpendicular to it. Calculate the torque produced.	3	OR	State the principle of moments . A metre rule is pivoted at its centre. A 4 N weight hangs 30 cm to the left of the pivot. At what distance on the right must a 6 N weight hang to balance it?	3
(vii)	Define centripetal force and write its formula. A 0.5 kg stone is whirled in a horizontal circle of radius 1 m at a speed of 4 m/s. Calculate the centripetal force.	3	OR	Define orbital velocity . A satellite orbits at a radius of 4.2×10^7 m and completes one orbit in 86,400 s. Find its orbital speed ($v = 2\pi r/T$).	3
(viii)	State Hooke's law and define the spring constant with its SI unit. A spring stretches by 4 cm when a 12 N load is hung from it. Calculate its spring constant.	3	OR	Sketch a force-extension graph for a spring obeying Hooke's law and mark the elastic limit . State what the slope of the straight portion represents and what happens beyond the elastic limit.	3
(ix)	Define pressure and write its SI unit. A box weighing 600 N rests on the floor with a base area of 1.5 m ² . Calculate the pressure it exerts on the floor.	3	OR	Calculate the pressure exerted by a column of water 3 m deep. (density of water = 1000 kg/m ³ , $g = 10$ m/s ²). State the two factors on which liquid pressure depends.	3
(x)	Define work and state its SI unit. A boy does 250 J of work pushing a cart 10 m along a level floor with a force along the motion. Calculate the force he applied.	3	OR	Define power and state its SI unit. A pump does 6000 J of useful work in 30 s. Calculate its power output.	3
(xi)	List, in order, the main steps of the scientific method used to investigate a natural problem.	3	OR	What is the role of physics in the development of technology? Give two examples where a discovery in physics led to a useful technology.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Define acceleration . A motorcycle accelerates uniformly from 10 m/s to 30 m/s in 8 s. (a) Find its acceleration. (b) Find the distance it covers in this time. (c) State what the area under its velocity-time graph represents.	5	OR	Write the three equations of motion for uniform acceleration and define every symbol. A ball is dropped from rest and hits the ground after 3 s. Find (a) its velocity at impact and (b) the height of the fall. ($g = 10$ m/s ²)	5
3(ii)	Define momentum and state the law of conservation of momentum . A 1500 kg car moving at 12 m/s collides with a stationary 1000 kg car and the two lock together. Find their common velocity just after the collision.	5	OR	Define momentum . Using the law of conservation of momentum, explain how a rocket is pushed forward by ejecting hot gases backward. Why does the rocket speed up as it burns more fuel?	5
3(iii)	State Pascal's principle and explain the working of a hydraulic lift. A hydraulic lift has a small piston of area 0.01 m ² and a large piston of area 0.5 m ² . If a force of 40 N is applied to the small piston, find the force produced on the large piston.	5	OR	Derive the formula $P = \rho gh$ for the pressure of a liquid column. Using it, find the total pressure at the bottom of a 5 m deep water tank. (atmospheric pressure = 1.0×10^5 Pa, density of water = 1000 kg/m ³ , $g = 10$ m/s ²)	5
3(iv)	Define power and efficiency , writing the formula for each. A water pump takes in 5000 J of electrical energy and delivers 4000 J of useful work in 25 s. Calculate (a) its efficiency and (b) its useful power output. (c) Explain why no real machine can be 100% efficient.	5	OR	State the law of conservation of energy . A 3 kg ball is dropped from a height of 5 m. Find (a) its potential energy at the top, (b) its kinetic energy just before hitting the ground, and (c) its speed at the ground. ($g = 10$ m/s ² , ignore air resistance)	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Units 1-6 & 9 — Measurement, Kinematics, Dynamics-I, Dynamics-II, Pressure & Deformation, Work & Energy (till 6.4), Nature of Science

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